Missing String

LIMITS `2s, 512 MB` · Interactive



This is an **Interactive Problem**.

The judge has a secret binary string S of length $n (1 \leq n \leq 1000)$.

You can do several queries on the string.

To do a query, print "? Q". Here Q is any binary string of length $m (1 \leq m \leq 2 \times n)$.

The judge replies to your query with an integer, the number of substrings of S that are lexicographically smaller than Q.

Find the secret string by asking no more than $2 \times n$ queries.

When you have figured out the secret string, print "! S" in a line and move on to the next test case.

INPUT

The first line contains a single integer $T (1 \leq T \leq 100)$ — the number of test cases.

Each test case starts with reading the integer n, the length of the secret string in that test case. The sum of n in all the test cases does not exceed 2000.

Interaction Details

For each query, print the query in a line and flush the output. The query should be in the format mentioned above.

After each query, read an integer: the response from the judge. -1 as a response from the judge means you did an invalid query. If you do any invalid query, your submission will fail with an arbitrary incorrect verdict.

After confirming the secret string in a test case, move on to the next test case if there is any, terminate your program otherwise.

To flush the output stream, you may use:

- `fflush(stdout)` in C/C++
- `stdout.flush()` in Python

OUTPUT

A sample interaction is given below for a test with two test cases, the first string is “001” and the second string is “11”.

```
>> 2
>> 3
<< ? 0
>> 0
<< ? 1
>> 5
<< ! 001
>> 2
<< ? 0
>> 0
<< ? 1
>> 0
<< ! 11
```

Another sample with a single test case and the secret string being “01”.

```
>> 1
>> 2
```

```
<< ? 0101
>> 2
<< ? 101011
>> -1
```

Here `>>` indicates what your program reads and `<<` indicates what your program writes. These symbols are here to make it easy to understand. You do not have to print such symbols from your program.

NOTES

Notes

- Binary strings are strings consisting of characters 0 and/or 1.
- Substrings of a string are collections of one or more consecutive characters of that string in the same order they appear in the original string.
- String A is said to be lexicographically smaller than string B if either string A is a prefix of string B or the first index where characters in the strings mismatch, character in string A has a lower ASCII value.

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Number of Anagrams

LIMITS `2.5s, 512 MB`

You will be given a list L. Each element L_i will contain a list of integers. You have to perform two types of queries in list L.

- $1\ i\ l\ r\ x\ y$

  This is the query of type 1 which asks you to replace the elements $L_{i_k} (l \leq k \leq r)$ with y if $L_{i_k} = x$.

- $2\ i$

  Query type 2 asks you to determine the number of anagrams of L_i in list L.

Definitions:

- **Anagram**: Two arrays are anagrams if one can be formed by rearranging the elements of the other. Note that, in this problem, we consider an array to be an anagram of itself.

INPUT

The first line of the input file will contain a positive integer $N (1 \leq N \leq 10^5)$ which denotes the number of elements in list L.

Each of the next N lines will start with a positive integer M denoting the number of elements of L_i. Following by a space there will be M space separated positive integers. Note that, $(1 \leq L_{i_j} \leq 10^4)$.

Then, you will be given a positive integer $Q (1 \leq Q \leq 6 \times 10^5)$ in a new line.

After that you will be given the queries in the next Q lines as described in the statement.

- $1\ i\ l\ r\ x\ y$

Here, $(1 \leq i \leq N)$, $(1 \leq l \leq r \leq |L_i|)$, $(1 \leq x, y \leq 10^4)$

- $2\ i$

  Here, $(1 \leq i \leq N)$

Note that, $(1 \leq \sum_{i=1}^{N} |L_i| \leq 10^5)$ and there will be at least one query of type 2.

OUTPUT

For each query of type 2, print the answer in a new line.

EXAMPLE

INPUT	OUTPUT
3 2 10 20 2 20 10 2 10 10 5 2 2 2 3 1 1 2 2 20 10 2 2 2 3	2 1 1 2

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Meena, Raju and Mango Tree

LIMITS `1s, 512 MB`

Clever girl Meena likes to climb trees. One day her brother Raju, after coming back from school, said to Meena with joyful eyes, “I have seen a big tree with many ripe mangoes. Let's go climb that one!”.

We can think of the mango tree as a connected undirected graph with no cycles and having exactly one path from one node to another. Meena asked Raju to describe the tree. Raju could not memorize the exact structure of the tree but remembered how many nodes were at each level considering the tree is rooted. Suddenly, they both got worried as many trees could satisfy that condition and Meena could not climb trees with all diameters.

In tree data structures, the root node is said to be at level 0, the root node's children are at level 1, and the children of that node at level 1 will be at level 2, and so on. And, the diameter of a tree means the number of nodes on the longest path in the tree.

Can you help Meena and Raju determine how many unique diameter lengths are possible for any such tree?

Suppose that, the mango tree has n levels. For each level i $(0 \leq i < n)$ there exists $a[i]$ nodes, 0-indexed.
You will be given an array b of length m $(m < n)$, 1-indexed.
Now, to construct a from b, you need to follow the following instructions:

- $a[0] = 1$
- $a[i] = b[i]$ for $1 \leq i \leq m$
- $a[i] = a[i - m]$ for $m < i < n$

INPUT

The first line will contain T $(1 \leq T \leq 10^5)$, the number of test cases.
For each test cases, the first line will contain two integers $n,\ m$ $(2 \leq n \leq 10^9,\ 1 \leq m < min(n, 10^5))$, the number of levels in the tree and the length of the array b.
In the next line, there will be m integers denoting the elements of array b $(1 \leq b[j] \leq 10^9, for\ all\ 1 \leq j \leq m)$.

It's guaranteed that,

$\sum m$ over all test cases $\leq 2 \times 10^5$.

OUTPUT

For each test case, output a single integer denoting the possible number of unique diameter lengths.

EXAMPLE

INPUT	OUTPUT
1 3 1 2	2

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Pop! Shop

LIMITS `1s, 32 MB`

In a bustling medieval town, Arwa stumbled upon a lively carnival, its tents and banners fluttering in the breeze. Amidst the crowd's chatter and laughter, she spotted a peculiar stall adorned with colorful balloons and mysterious prizes. Curiosity piqued, she approached the stall, where a wizened carnival master awaited. "Welcome, fair maiden, to the Pop! Shop" he proclaimed, gesturing towards a row of N balloons, each bearing its own enigmatic value, v_i.

"In this grand spectacle, ye shall wield special darts, each endowed with magical properties.", the carnival master explained, "With these darts, ye can perform daring feats to pop the balloons and earn points." He continued, "Be wary maiden, you can leave a balloon untouched, but each balloon can be popped only once. And the more points ye amass, the grander the reward!". With a twinkle in her eye, Arwa listened intently as the carnival master detailed the ways she could wield her dart:

- **The Single Shot:** Arwa could aim her dart at a single balloon, popping it to earn its value v_i;
- **The Double Strike:** By targetting two adjacent balloons simultaneously - the i-th and $(i+1)$ -th balloons - Arwa's magical dart, when thrown, would unleash double darts popping the two balloons, earning the product of their values $v_i \times v_{i+1}$;
- **The Spectacular Sweep:** Should Arwa choose, she could aim her magical dart in a way that hits two different ballons i and $j > i+1$, earning the value $v_i \times v_j$. But she needs to keep two things in mind when using this move:
    1. Sweep on balloons i and j can be performed only if all balloons between them, from $(i+1)$ to $(j-1)$, had been popped earlier.
    2. In case she decides to perform multiple sweeps, their ranges cannot be disjoint. Given Sweep 2 (involving balloons i_2 and j_2) is performed after Sweep 1 (involving balloons i_1 and j_1), Arwa needs to maintain $i_1 > i_2$ and $j_1 < j_2$. This should be followed for all subsequent cases of sweeps.

Eager to embark on this daring adventure, Arwa faced the challenge: given the number of balloons and their associated values, she sought to devise a strategy to maximize her score. With determination in her heart, Arwa pondered her options, ready to seize the opportunity and claim the greatest prize the carnival had to offer.

INPUT

The first line of the input contains a single integer $N(1 \leq N \leq 5000)$ denoting the number of balloons.

The following line contains N space-separated integers, $v_i(-10^7 \leq v_i \leq 10^7)$ indicating the value associated with each balloon.

OUTPUT

Print a single integer denoting the maximum score.

EXAMPLES

INPUT	OUTPUT
3 4 5 3	23

Arwa can use the Double Strike to pop the first two balloons getting $4 \times 5 = 20$ points. Then she can use the Single Shot to pop the third balloon to get 3, totalling 23 points.

INPUT	OUTPUT
7 9 5 15 20 13 17 30	932

Arwa can perform a Single Shot to get 5 and use the Double Strike to get 15×20. Then she can perform the Spectacular Sweep to get 9×13 and finally use the Double Strike to

get 17×30. Note that, if she were allowed to perform disjoint sweeps, she could get $5 + 9 \times 15 + 13 \times 17 + 20 \times 30 = 961$, but the carnival master will not allow it.

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Ri, the Miner

LIMITS `2s, 512 MB`

Gondor has n circular gold mines with IDs from 1 to n containing some gold. Each mine's center can be located by a 2D coordinate.

Ri, the miner, lives in a place in Gondor that a coordinate on a 2D plane can locate. She can dig mines that only cover her place. In other words, she can dig a mine if her location lies inside or on the boundary of that mine.

Even though it would have been better to dig all the mines, she unable to move much for sickness and can only dig one mine at most. So, she wants to dig the mine that contains the most amount of gold. Help her to find the most suitable one.

INPUT

The first line of the input will consist of an integer t $(1 \leq t \leq 10^5)$— the number of testcases. The description of each testcase follows:

The first line of the testcase will consist of an integer n $(1 \leq n \leq 3 \cdot 10^5)$ — the number of gold mines.

Next n lines will contain four integers, $M_{x_i},\ M_{y_i},\ r_i,\ a_i (-10^9 \leq M_{x_i}, M_{y_i} \leq 10^9, 1 \leq r_i,\ a_i \leq 10^9)$— the coordinates of the center of the mines, the radius of the mines, and amount of gold in each mine respectively.

Next line will contain an integer q $(1 \leq q \leq 3 \cdot 10^5)$— number of queries.

Each query will contain two integers in a line, x_i, y_i $(-10^9 \leq x_i, y_i \leq 10^9)$— coordinates of Ri's location.

Additional constraints for the queries: Either $x_i - x_j = 0$ or $y_i - y_j = 0$ or both for all $1 \leq i < j \leq q$.

Moreover, you can assume that the sum of n and the sum of q over all testcases will not exceed $3 \cdot 10^5$ respectively.

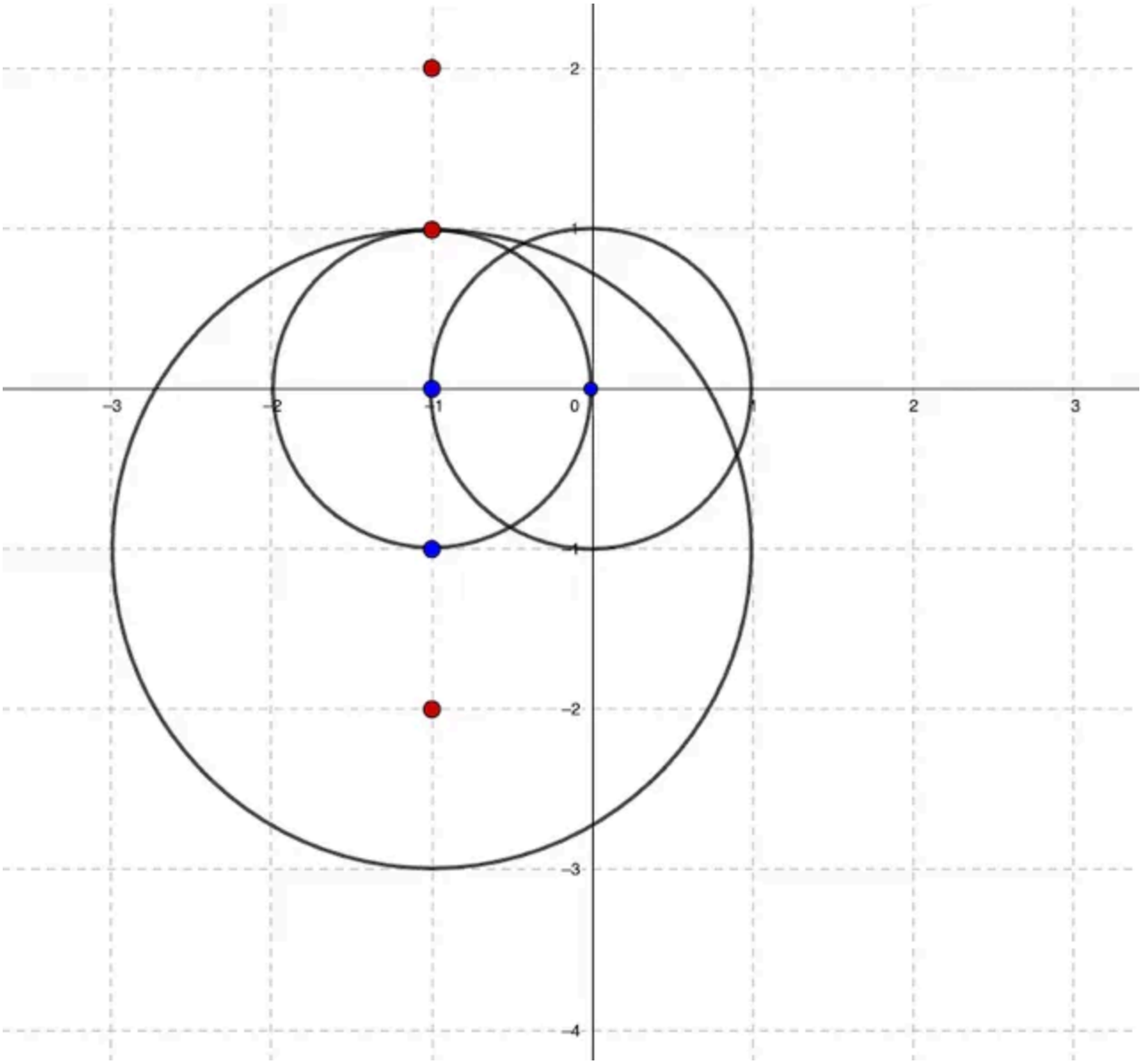
OUTPUT

For each query, print the ID of the mine that Ri will dig. If there are multiple mines, print the minimum ID. If no mines cover Ri's place, print -1.

EXAMPLE

INPUT	OUTPUT
1 3 0 0 1 3 -1 0 1 10 -1 -1 2 9 3 -1 -2 -1 1 -1 2	3 2 -1

See the image to understand the sample case. Here, the $blue$ dots are the center of the circles and the red dots are the places of Ri.



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Efficient Tour

LIMITS `1s, 512 MB`

Gopher is embarking on yet another adventurous tour, this time across N cities, each $uniquely$ defined by its two-dimensional coordinates and having $distinct$ Power. Gopher will visit all the cities but can only visit each city once. He will start from the city with the lowest Power and return to the starting city after visiting all the cities.

But, as with any journey, there are rules to abide by:

1. Before reaching the city with the highest Power, Gopher can go from city i to city j only if the Power of city i is lower than that of city j $(Power_i < Power_j)$.
2. Once he reaches the city with the highest Power, Gopher's path can go from city i to city j if the Power of city i is higher than that of city j $(Power_i > Power_j)$.

The cost of moving from city i to city j is calculated using the formula $(x_i - x_j)^2 + (y_i - y_j)^2$, where (x_i, y_i) represents the coordinates of city i and (x_j, y_j) represents the coordinates of city j.

Gopher could complete his tour through various routes, he wants to accomplish it with the minimum cost. Thus, he turns to you for assistance in this.

INPUT

The first line will contain one integer $N(2 \leq N \leq 3 \times 10^3)$ denoting the number of cities Gopher will visit in his tour.

Next N lines will contain three integers $Power_i(1 \leq Power_i \leq N)$, $x_i, y_i(1 \leq x_i, y_i \leq 10^6)$, $Power_i$ denoting the Power of i_{th} city and x_i, y_i denoting the coordinator of i_{th} city.

OUTPUT

Print a single line containing the minimum cost to visit all the cities in the Gophers tour. Errors less than 10^{-4} will be ignored.

EXAMPLE

INPUT	OUTPUT
4 1 2 3 2 6 3 3 3 7 4 5 2	14.9225

The minimum cost path is (2,3)->(6,3)->(5,2)->(3,7)->(2,3).

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Name Game

LIMITS `1s, 256 MB`

> *React anything on this post. I will give you three names. And you have to comment about them without mentioning their names.*

We all played this trendy game on Facebook once upon a time. Now, we will play this game again.

> *Participate in this contest. I will give you* some *names. And you have to print something about them without mentioning the name.*

As you have already participated, I will give you some names and you have to print a line of strings about them without revealing their names. In other words, you can print anything about them but you can not mention the name, not even as a substring.

INPUT

The first line of the input is an integer, t $(1 \leq t \leq 1000)$— denoting the number of names that I will give you.

Next t lines will contain a non-empty string denoting a name. The name will consist of Latin letters and will have at most 100 characters.

OUTPUT

For each testcase, print a line containing something about the name within 10000 characters without mentioning the name, not even as a substring. To avoid inconsistency, you are allowed to print only Latin letters and whitespace.

Note that, names are case-insensitive and you should not print only whitespace(s) or blank string cause printing nothing is a violation of the game's rules.

EXAMPLE

INPUT	OUTPUT
4 Ri rafi sumaiya sakib	sweet Girl he is nerd she is nerd Osthir Balok

For the 4th test case, you can not print `sakib osthir balok` or `saKibbai shera` because these statements reveal the name `sakib`. But you can print `boss player` as it does not reveal the name.

NOTES

Note

A string t will be called a substring of s if t can be obtained by removing some (possibly zero or all) characters from s from the beginning or from the end or both. For example, the list substrings of the string "*apple*" would be "*apple*", "*appl*", "*pple*", "*app*", "*ple*", "*le*", "*a*", "*p*", "*l*", "*e*", ""(empty string) . But "*apl*", "*pe*", "*ep*" are not substrings of "*apple*".

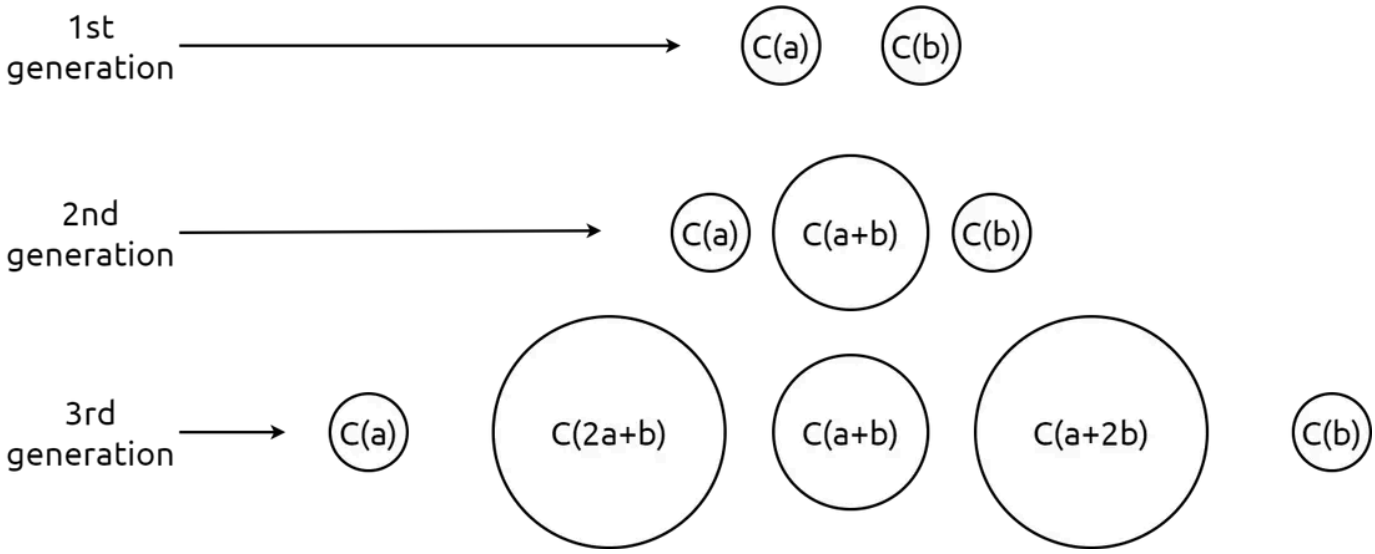
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Doctor Strange in the Circleverse

LIMITS 1s, 512 MB

At the beginning of time, there were only two circles in the circleverse (a universe where everyone is a circle). They are called the first generation of the circleverse and known as $C(a)$ and $C(b)$, where $C(r)$ denotes a circle with radius r.

Each year, a new generation of the circleverse is created from the previous one. Each generation of the circleverse consists of all circles in it positioned side by side on a straight line. To create the new generation, each pair of adjacent circles of the previous generation creates a new circle and positions it between them. The new circle gets a radius equal to the sum of the two adjacent circles it is created from. The figure below shows the first three generations of a circleverse.



Doctor Strange, the master of mystic arts, has explored the multiverse and found two circleverses C_1 and C_2. First generation of C_1 consists of circles $C(0)$ and $C(1)$. On the other hand, both circles of the first generation of C_2 are $C(1)$. Doctor Strange wants to use C_1 and C_2 to create a new generation of circles he calls super-circles.

Let,
$C(x)$ = ith circle of the nth generation of C_1
$C(y)$ = ith circle of the nth generation of C_2

P = number of circles in the nth generation of a circleverse

Doctor Strange will create P super-circles initially. The ith super-circle will be created from $C(x)$ and $C(y)$ and it will be $C(x+y)$. After that, Doctor Strange will destroy all unstable super-circles. A super-circle $C(x+y)$ is unstable if $x > y$ or $y > n$. To destroy these unstable super-circles, he needs to summon enough energy from the multiverse. The amount of the energy required depends on the number of unstable super-circles.

Since Doctor Strange is the master of mystic arts, not mathematics, he needs your help to calculate the number of unstable super-circles. Given n, calculate the number of the unstable super-circles Doctor Strange will have to destroy.

INPUT

First line of the input will contain an integer T, the number of test cases.
Each of the next T lines will contain an integer n, indicating that circles from the nth generation will be used to create super-circles.

$1 \leq T \leq 10^5$
$1 \leq n \leq 10^7$

OUTPUT

For each test case, print the number of the unstable super-circles modulo $10^9 + 7$ in a single line.

EXAMPLE

INPUT	OUTPUT
5 1 3 5	0 0 6 2002 988182602

```
12
100
```


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Prime - Prime - Prime

LIMITS `1s, 512 MB`

Given an integer n, print n **unique** integers in the range $[1, 10^9]$ such that,

1. The bitwise AND value of all the numbers is a prime number.
2. The bitwise OR value of all the numbers is a prime number.
3. The bitwise XOR value of all the numbers is a prime number.

INPUT

Input consists of a single integer $n (1 \leq n \leq 10^5)$.

OUTPUT

Print n space-separated integers in a line. Do not print any extra spaces after the last integer.

EXAMPLE

INPUT	OUTPUT
2	2 7

NOTES

Notes

- A ***prime number*** is a natural number greater than 1 and has exactly 2 divisors which are 1 and the number itself.

- ***Bitwise OR*** operation compares bits in the same index in the binary representation of all the numbers in question and if at least one of the bits was 1, it sets the corresponding resulting bit to be 1, otherwise 0.
- Similarly, ***Bitwise AND*** operation sets the corresponding bit to 1 if all the bits were 1, otherwise 0.
- ***Bitwise XOR*** operation sets the corresponding resulting bit to 1 if an odd number of bits were set to 1, otherwise 0.

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Thanos LLC.

LIMITS `1s, 512 MB`

On the planet Titan, there are N residents, numbered from 1 to N, all employed by Thanos. For each i from 1 to N, i^{th} person's salary can be described as S_i.

On planet Titan, a network of people can be described as an unweighted directed graph containing N nodes. Each node represents a person and each edge represents a one-way friendship. There can be duplicate edges. In this network, individuals who share mutual friendships are considered part of the same community. These communities form groups where each member can reach every other member through a series of interconnected friendships.

You need to execute Q queries on this graph, each falling into one of three categories:

- $AC\ X\ Y$: Add amount X to the income of each person within the community to which person Y belongs.
- $AS\ X\ Y$: Add amount X to the income of person Y directly.
- Q: Determine the maximum individual income across the entire planet and identify the maximum number of people sharing that income within a single community.

Note that the value of the amount X can be either positive or negative. If an individual's income becomes negative, it indicates they are in debt to Thanos. In case of all negative income in the whole planet, print a single integer 0, otherwise print two integers indicating the maximum individual income across the planet and the maximum number of people who earn that amount within a single community.

INPUT

The first line of input consists of three integers: N, M, and Q $(2 \leq N, M, Q \leq 2 \times 10^5)$, representing the number of nodes, the number of edges, and the number of queries to perform, respectively.

Each of the next M lines contains two integers, U and V $(1 \leq U, V \leq N, U \neq V)$, indicating a directed edge from node U to V.

The next line contains N space-separated integers, denoted as S_i $(1 \leq |S_i| \leq 10^9)$ for each i from 1 to N, representing the salary of the ith person.

The following Q lines each contain one of three types of queries, presented in the following format:

- $AC\ X\ Y$ $(1 \leq Y \leq N, 1 \leq |X| \leq 10^9)$
- $AS\ X\ Y$ $(1 \leq Y \leq N, 1 \leq |X| \leq 10^9)$
- Q

OUTPUT

For each of the Q query types, print two integers S and P on a single line separated by a space. S represents the maximum individual income across the planet, and P represents the maximum number of people sharing that income within a single community. In case of all negative income in the whole planet, print a single integer 0.

EXAMPLE

INPUT	OUTPUT
6 6 16 1 2 2 3 2 4 2 5 5 2 3 6 5 2 3 1 6 10 Q AC 5 2 Q AS 4 2	10 1 11 1 11 2 11 2 15 1 0

```
Q
AS 1 6
Q
AS 10 1
Q
AS -100 2
AC -100 1
AC -100 4
AC -100 3
AC -100 6
AC -100 5
Q
```


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Fuel Finance

LIMITS `2s, 512 MB`

X Game Studio is working on their next super-hit game. The game is quite simple. The player has a car and there are n bridges in front of them. The player has to use their car to cross each bridge once, they can, however, choose in what order to cross the bridges. The ith bridge has a fuel usage value f_i and a fuel refill value r_i. When the player's car crosses the bridge, the car loses f_i units of fuel, but at the end of the bridge, the car gets additional r_i amounts of fuel. Of course, if the fuel ever becomes negative, the car will stop and the player will lose. The player has one chance to initially fill the car's fuel by some amount. So the player needs to do some calculations to know how much fuel to keep in the beginning and in what order to cross the bridges. If they keep too little fuel, the car will stop on some bridge; if they keep too much, they waste their in-game currency.

The developers found that this was too easy, so they added some more mechanics. The bridges can have some locks at their entrance. In particular, the ith bridge has a lock with the color l_i in its entrance. If $l_i$is 0, then the bridge has no lock. Also, the ith bridge has a key with the color k_i at the end of the bridge. Here, k_i being 0 means the bridge has no key. **Note that multiple locks might have the same color, but for each color, there is at most one key.** This lock-key mechanic creates more restrictions in the game, as to open a lock, player must have the corresponding key (keys can be used multiple times). Now happy with the rules, the developers have made the levels so that it's possible to finish the level. But now they need to know the minimum amount of fuel the player needs to fill initially to cross all the bridges as this value can be used to quickly give verdicts instead of making players fumble around with bridges. This is where you come in, The level description is given. You have to find the minimum initial fuel needed to make sure the player can finish the game.

INPUT

The first line of input will have a positive integer T, the number of independent test cases. Each test case will begin with a positive number n, the number of bridges. Then the next n lines each

will have 4 space-separated values denoting f_i, r_i, l_i, k_i.

- $1 \leq T \leq 10^5$
- $1 \leq n \leq 10^5$
- $1 \leq f_i, r_i \leq 10^5$
- $0 \leq l_i, k_i \leq 10^5$
- On a level, no two bridges have the same positive k_i.
- It is guaranteed that there is some sequence of bridges using which the player can finish the level.
- The sum of n over all test cases $\leq 5 \times 10^5$

OUTPUT

For each testcase, output the minimum initial amount of fuel the player needs to fill initially to finish the game in a separate line.

EXAMPLE

INPUT	OUTPUT
1 4 2 5 0 1 3 3 1 0 4 2 1 2 10 5 2 0	9

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